

Random Forest Regression

Import libraries

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Import the dataset

```
In [3]: df = pd.read_csv("01_dataset/Position_Salaries.csv")
X = df.iloc[:,1:-1].values
y = df.iloc[:, -1].values
```

Training the Random Forest Regression Model on the whole dataset

```
In [4]: from sklearn.ensemble import RandomForestRegressor
regressor = RandomForestRegressor(n_estimators = 10, random_state = 0)
regressor.fit(X, y)
```

Out[4]:  RandomForestRegressor

► Parameters

Predicting a new result

```
In [5]: regressor.predict([[6.5]])
```

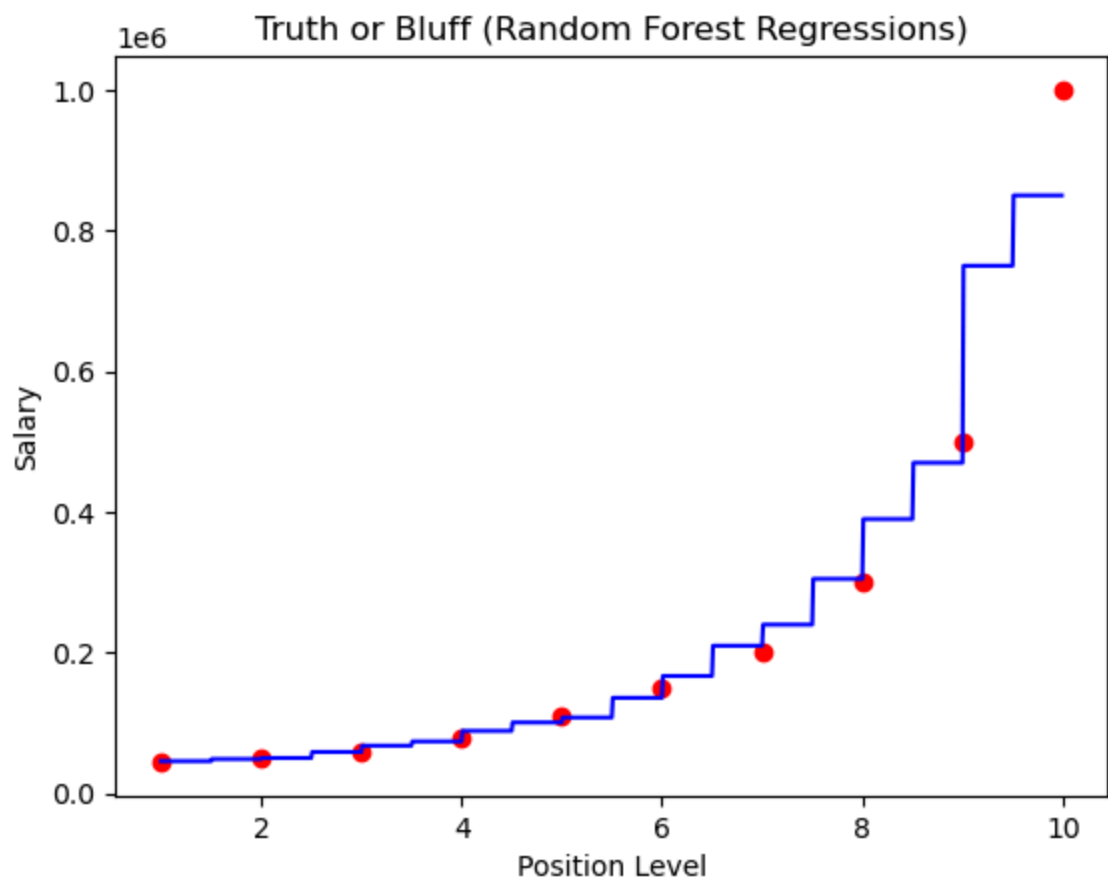
Out[5]: array([167000.])

Visualising the Random Forest Regression results(higher resolution)

```
In [6]: X_grid = np.arange(min(X), max(X), 0.01)
X_grid = X_grid.reshape(len(X_grid), 1)
plt.scatter(X, y, color = 'red')
plt.plot(X_grid, regressor.predict(X_grid), color = 'blue')
plt.title("Truth or Bluff (Random Forest Regressions)")
plt.xlabel('Position Level')
plt.ylabel('Salary')
plt.show()
```

/tmp/ipykernel_20279/1265177303.py:1: DeprecationWarning: Conversion of an array with ndim > 0 to a scalar is deprecated, and will error in future. Ensure you extract a single element from your array before performing this operation. (Deprecated NumPy 1.25.)

```
X_grid = np.arange(min(X), max(X), 0.01)
```



In []: